The unit economics of catching theft pay for themselves in weeks.

A B2G venture that recovers stolen electricity for Egypt's distribution companies — hardware on the transformer, AI in the cloud, and a share of every kilowatt it puts back on the bill.

EGP 48 = $1	~230,000	EGP 25–35B	< 2 months
reference rate (2026)	distribution transformers (est.)	annual non-technical loss	utility payback per node

01 BUSINESS MODEL

How the money flows

CUSTOMER SEGMENTS

- Distribution companies (EDCs) — primary
- Ministry of Electricity / regulator
- Industrial estates; MENA utilities next

VALUE PROPOSITION

- Recover stolen energy — new revenue, not tariffs
- Separate theft from natural loss → few false alarms
- Explainable verdict an inspector acts on

REVENUE STREAMS

- Hardware sale / HaaS lease
- SaaS platform subscription
- Performance / recovery share

KEY RESOURCES · MOAT

- Labelled theft dataset (compounds)
- Edge hardware + detection IP
- Ministry & field-ops relationships

CHANNELS

- Direct BD to EDCs & Ministry
- Paid pilots → framework tenders
- System integrators

COST STRUCTURE

- Hardware COGS & logistics
- Cloud + ML compute
- Field deployment & connectivity

02 REVENUE

Three revenue streams

Lead with a hardware sale + a light SaaS fee, then capture upside on the energy we actually recover.

STREAM 01 · one-time

Hardware

EGP 6,000 / node

Edge node + gateway share at ~25% margin. Or HaaS lease to shift CapEx off the utility.

STREAM 02 · recurring

SaaS platform

EGP 300 / node / month

Detection, live map, explainable verdicts, reporting. ~75% gross margin.

STREAM 03 · performance

Recovery share

15% of recovered energy

A cut of verified recovered billing for 24 months per zone. We win only when they do.

Every part, to spec

A low-cost prototype proves the physics; a kiosk-grade production node ships to the transformer.

PROTOTYPE · PROOF OF CONCEPT

ESP32 DevKit	EGP 250
SCT-013-000 CT clamp	EGP 350
Burden + bias circuit	EGP 60
ZMPT101B voltage sensor	EGP 150
Breadboard, PSU, jumpers	EGP 190
Total prototype	**≈ EGP 1,000**

PRODUCTION NODE (hardware)

Rogowski coil ×3	EGP 1,800
Metering AFE + MCU	EGP 600
LoRaWAN module (SX1262)	EGP 400
IP65 enclosure + power	EGP 1,200
Assembly, test & calibration	EGP 500
+ install EGP 1,500 · all-in	**≈ EGP 6,000**

One instrumented transformer ≈ EGP 6,000 (~$125) — and it watches a whole feeder of customers.

04 UNIT ECONOMICS

One node, five-year life

All-in cost to deploy (CapEx)	EGP 6,000
Hardware gross margin (one-time)	EGP 1,500
SaaS revenue / year	EGP 3,600
Recovery-share / year (blended)	EGP 1,900
Recurring gross profit / year	EGP 3,850
Allocated CAC (acq. + install)	EGP 2,500
5-yr gross profit / node (LTV)	**≈ EGP 20,750**
LTV / CAC	**≈ 8×**

THE CUSTOMER'S MATH

EGP 3,500 / mo

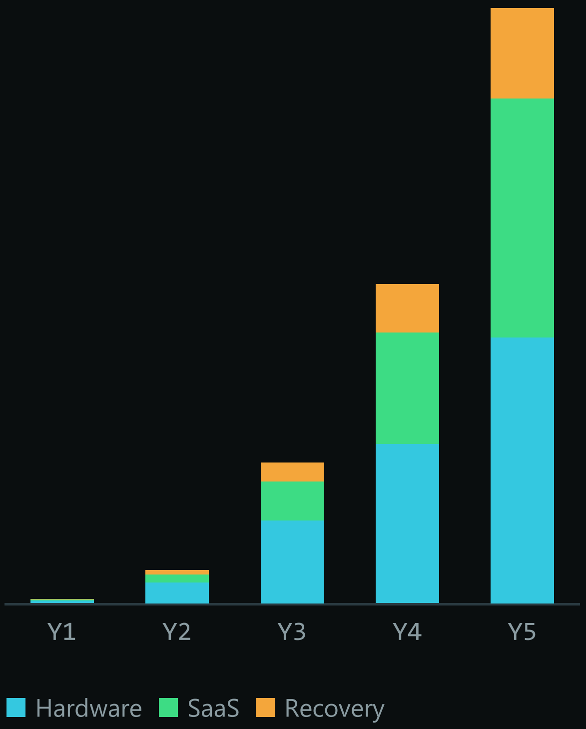
recovered when a node catches one continuous ~3 kW illegal draw (≈2,160 kWh/mo).

Against an EGP 6,000 node — **payback in under two months.**

Five-year model — EGP millions

	Y1	Y2	Y3	Y4	Y5
Total revenue	4.4	33.5	142.8	324.4	604.8
Gross profit	1.9	15.0	67.0	167.1	327.8
Operating expenses	12.0	28.0	50.0	95.0	160.0
EBITDA	(10.1)	(13.0)	17.0	72.1	167.8
Transformers (k)	0.5	4	18	45	90

Revenue by stream



Breakeven in Year 3 as recurring revenue overtakes the cost base.

Onto the real iron box

The field target is a sealed distribution kiosk (RMU) in Kafr El-Sheikh — retrofitted non-invasively on the LV side.

Distribution kiosk (RMU) · Kafr El-Sheikh

- ✓ **Wrap 3 Rogowski coils**
  around the LV busbars — one per phase, no cables cut
- ✓ **IP65 edge node**
  custom PCB in the panel; antenna outside the metal shell
- ✓ **MV chamber never touched**
  magnetic coupling — no galvanic contact, no outage
- ✓ **Scales to 90,000 kiosks**
  ~30 min, reversible, tamper-resistant per unit

Where we sit

Six ways utilities fight theft today — only one is low-CapEx, explainable, and works without a smart-meter rollout.

APPROACH	Needs AMI	Sep. theft	Explains	CapEx	Legacy grid
Manual field audits	No	✗	partial	labour	✓
Full AMI rollout	is meter	analytics	partial	very high	new only
AMI / MDM analytics	Yes	✓	✓	high	needs AMI
LV feeder monitoring	No	✗	✗	high	✓
Fixed-threshold alarms	No	✗	✗	low	✓
Grid-Pulse NTL	No	✓	✓	low	✓

The whitespace: low-CapEx + explainable + no AMI required — the corner every incumbent misses.

Why we win — and what could go wrong

THE MOAT vs ALTERNATIVES

Manual audits	random, slow
Smart-meter rollout	years · huge CapEx
Fixed-threshold analytics	floods on hot days
Grid-Pulse NTL	localized, explained, ROI < 2 mo

RISKS → MITIGATION

▲ Slow B2G procurement

paid pilot + audited recovery → tender

▲ Hardware CapEx at scale

asset-backed financing vs signed contracts

▲ Recovery-share decays

shift to recurring SaaS + new regions

▲ Model false positives

explainable verdicts + human-in-loop

EGP 30M seed · use of funds

EGP 30M

≈ $625K — carries the company through the district pilot and two years to the Year-3 EBITDA turn.

Field pilot — 1,000 nodes + gateways	EGP 12.0M
Product & ML engineering	EGP 7.5M
Hardware R&D + certification	EGP 4.5M
Team & G&A	EGP 3.6M
Go-to-market / BD	EGP 2.4M

Hardware at scale (Years 3–5) is funded by asset-backed financing against signed EDC contracts — keeping the equity raise small vs the EGP 600M+ revenue it unlocks.